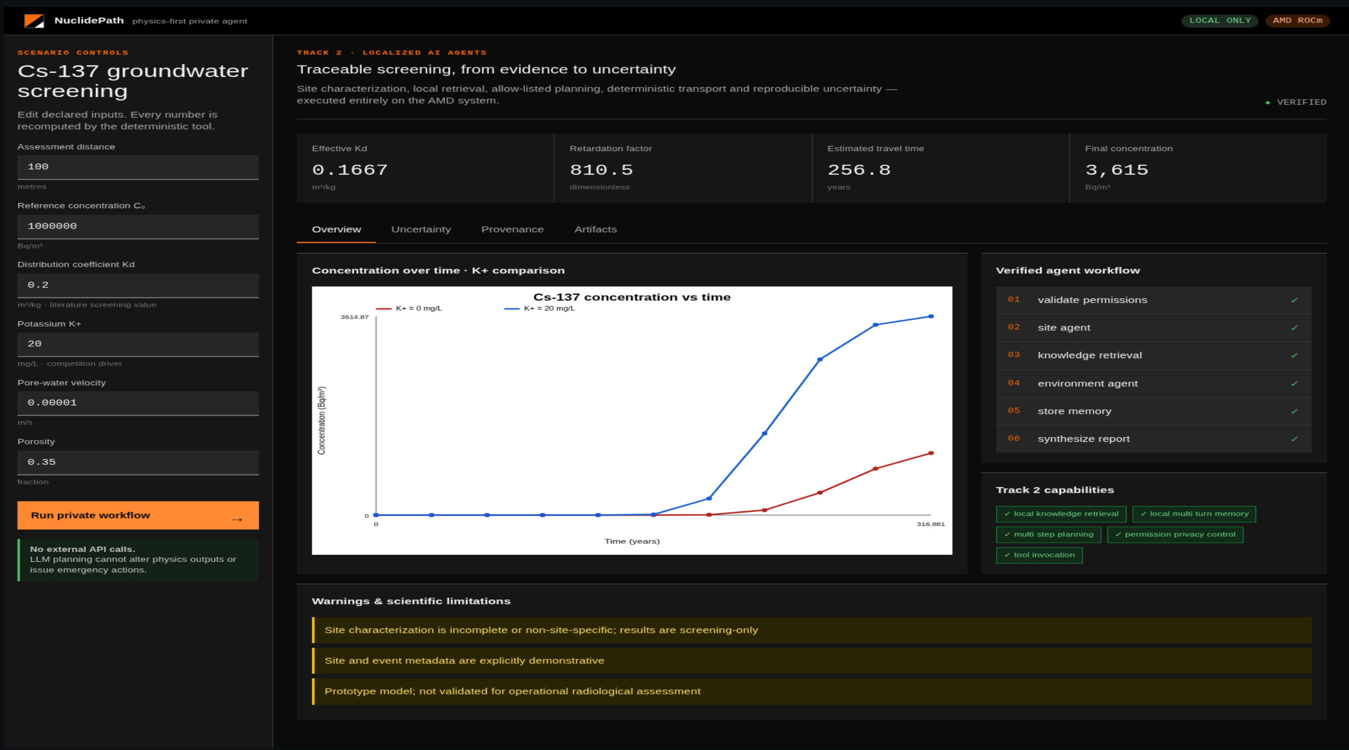
AMD AI DEVMASTER HACKATHON 2026 · TRACK 2

NuclidePath

Private agents. Deterministic physics.

A fully local, physics-first AI workflow for traceable Cs-137 groundwater screening.



5/5	231	356/15	0
Track 2 capabilities	AMD core snapshot	latest PR gate	cloud calls

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The optional PHREEQC multicomponent bridge is documented in the current source but is not shown in this core visual package; it remains PROCESS-QUALIFIED ONLY.

NuclidePath — Project Specification 1.0

Track: AMD AI DevMaster Hackathon 2026, Track 2

Team: Physics-First AI

Application: NuclidePath

Status: implemented current source snapshot; submission package and public contest sync remain gated

Last updated: 4 August 2026

1. Scenario and problem

Groundwater contaminant-screening workflows combine incomplete site data, scientific literature, numerical models and analyst judgement. A general-purpose LLM is useful for planning and explaining this workflow, but it is not a trustworthy calculator and must not invent physical values.

NuclidePath demonstrates a private local agent that answers a bounded question:

> For a hypothetical maintained Cs-137 boundary concentration in groundwater, how do sorption and dissolved potassium affect retardation, travel time and dissolved-concentration breakthrough at an assessment point?

The product is designed for transparent preliminary screening and education. It is not a regulatory model, dose code or emergency decision system.

2. Users

- nuclear/environmental engineers exploring screening assumptions;
- researchers studying contaminant transport and sorption;
- reviewers auditing an AI-assisted technical workflow;
- hackathon judges evaluating private agent deployment on AMD hardware.

3. Product modes

Dashboard

A localhost web interface exposes scenario inputs, deterministic outputs, agent steps, cited evidence, missing site data, uncertainty intervals and downloadable artifacts.

Full CLI

A composite JSON case runs the complete agent, retrieval, transport, sensitivity, uncertainty and reporting pipeline.

Offline validation

The deterministic planner exercises the same six workflow steps without an LLM. This is the CI/fallback path and proves that numerical outputs do not depend on generated text.

AMD-local LLM

Qwen3.5-9B Q8 runs through `llama.cpp`/HIP on the AMD GPU and returns only the allow-listed plan. It cannot modify scenario fields or tool results.

Optional PHREEQC chemistry bridge

The current source also exposes an opt-in, one-way compiler from an explicitly validated NuclidePath chemistry block to PHREEQC `SOLUTION`, `EXCHANGE`, `SELECTED_OUTPUT` and `TRANSPORT` input. Schema `nuclidepath-phreeqc-chemistry-2` accepts Cs plus one or more declared competitors from K, Na, Ca and Mg; K-free Na-Ca-Mg cases are supported. The compiler, dynamic parser, per-ion diagnostics, report bridge and ReplayBundle path are deterministic and CI-tested. The canonical transport model remains the authoritative screening result, and ordinary reports mark the external solver as compile-only/pending.

The bridge is `PROCESS-QUALIFIED ONLY`: the real Example 2 external execution is a process qualification record, while the paired Central Oklahoma cases still require a trusted run on `refs/heads/main`, a calibrated multi-cation Cs parameterization and an independent numerical/experimental oracle. No PHREEQC result is promoted to the canonical Kd path.

4. Track 2 capabilities

NuclidePath implements all five capabilities listed in the official rules.

4.1 Local knowledge retrieval

`LocalKnowledgeBase` indexes bundled Markdown documents and performs transparent TF-IDF ranking. A hit contains:

- document ID and title;
- matching excerpt;
- local path;
- source URL(s);
- numerical ranking score.

Retrieval requires no embedding service, cloud API or network connection.

4.2 Tool invocation

The Environment Agent calls `run_transport_contract`, which:

- validates and normalizes a strict JSON schema;
- constructs immutable physical parameters;
- invokes `simulate_transport` for every evaluation time;
- records derived values, assumptions, warnings and model version.

4.3 Multi-step planning

Both planners produce the same required sequence:

- `validate_permissions`;
- `site_agent`;
- `knowledge_retrieval`;
- `environment_agent`;
- `store_memory`;
- `synthesize_report`.

The orchestrator rejects unknown, missing or malformed steps.

4.4 Local multi-turn memory

`LocalMemoryStore` writes session-scoped JSONL with mode `0600`. It supports append, recent-context retrieval and explicit session deletion. Metadata is recursively redacted before persistence.

4.5 Permission and privacy control

`PermissionPolicy` defines role-specific allowlists. Default mode:

- permits validation, site assessment, local retrieval, physics, analysis, memory and reporting;
- denies external network access;
- always denies equipment control, operational control and public alerts;
- redacts names, emails, exact coordinates, tokens, API keys and passwords;
- binds the dashboard and LLM endpoint to loopback interfaces.

5. Functional requirements and evidence

ID	Requirement	Evidence
FR-01	Accept a complete structured scenario	`scenarios/cs137_emergency_full.json`
FR-02	Reject invalid types, ranges, unknown fields and non-finite values	contract/site/analytical tests
FR-03	Identify missing site characterization	`SiteAgent`, `test_site.py`

ID	Requirement	Evidence
FR-04	Retrieve local cited scientific context	`knowledge.py`, bundled corpus, retrieval tests
FR-05	Execute a safe multi-step plan	`workflow.py`, workflow tests
FR-06	Produce physical values only through deterministic tools	contract boundary and offline/LLM equality check
FR-07	Compare K+ competition cases	report runs at 0 and the scenario-selected K+ value (20 mg/L in the default case)
FR-08	Propagate declared parameter uncertainty	seeded Monte Carlo and quantiles
FR-09	Preserve private multi-turn context	mode-0600 JSONL memory and redaction tests
FR-10	Generate auditable outputs	JSON, Markdown, CSV, SVG and SHA-256 manifest
FR-11	Provide local CLI and dashboard	installable entry points and HTTP tests
FR-12	Run a private LLM on AMD	Qwen3.5-9B Q8 on ROCm/`gfx1100`
FR-13	Compile declared multicomponent chemistry to an external solver contract	PHREEQC schema 2, deterministic parser/diagnostics, report bridge and replay tests

6. Deterministic model

Effective sorption under empirical K+ competition

```
Kd_eff = Kd / (1 + alpha_K × [K+])
```

`alpha_K` is demonstrative unless experimentally calibrated.

Retardation and travel time

```
R = 1 + rho_b × Kd_eff / porosity
travel_time = distance × R / pore_water_velocity
```

Reactive advection–dispersion with a declared source

For `x >= 0`:

```
R ∂C/∂t = D ∂²C/∂x² − v ∂C/∂x − lambda R C
C(x,0)=0 for x>0 · C(0,t)=C0 · C(infinity,t)=0
A = sqrt(v² + 4 lambda R D)

C/C0 = 0.5 × [
  exp((v−A)x/(2D)) erfc((Rx−At)/(2 sqrt(DRt)))
  + exp((v+A)x/(2D)) erfc((Rx+At)/(2 sqrt(DRt)))
]
```

v is pore-water velocity and `C0` is a maintained boundary concentration. The solution reduces to classic Ogata–Banks when decay tends to zero, is bounded by `C0`, and is evaluated with a stable scaled-erfc implementation. The model version is `transport-prototype-0.3`.

7. Input contract

Required transport fields:

- `scenario_id`;
- `initial_concentration_bq_m3`;
- `distance_m`;
- `evaluation_times_s`;
- `distribution_coefficient_m3_kg`.

Optional fields have explicit defaults and units:

- `bulk_density_kg_m3`;
- `porosity`;
- `groundwater_velocity_m_s`;
- `dispersion_m2_s`;
- `potassium_mg_l`;
- `competition_coefficient_l_mg`;
- `half_life_years`.

Site context separately records event, radionuclide, pathway, description, provenance, classification and available characterization.

8. Parameter provenance and uncertainty

Every uncertainty range includes:

- lower and upper bound;
- distribution (`uniform`, `log_uniform` or `triangular`);
- classification (`site-measured-range`, `literature-range`, `demonstration-range`);
- human-readable source.

The bundled ranges intentionally mix one literature Cs Kd bracket with demonstrative hydraulic, dispersion, K+ and empirical-competition brackets. The report states that these inputs are independent and that reported intervals are not total predictive uncertainty.

9. Deployment

```
AMD Radeon Graphics · gfx1100
ROCm 7.2.1
llama.cpp Release HIP build
unsloth/Qwen3.5-9B-GGUF · Q8_0
```

```
8192-token context · all layers offloaded
OpenAI-compatible endpoint · 127.0.0.1:8000
Dashboard · 127.0.0.1:8080
```

No cloud inference is required. Python execution has no mandatory third-party runtime dependency.

10. AMD optimization plan and measured evidence

Implemented:

- HIP build targeted to `gfx1100`;
- Q8_0 selected because available VRAM permitted a conservative quantization;
- `-ngl 99` offloads all model layers;
- reasoning mode disabled for a short deterministic planning response;
- context limited to 8192 tokens;
- endpoint and dashboard restricted to localhost.

Measured AMD `transport-prototype-0.2` runtime checkpoint:

- prompt processing: `2861.56 ± 178.91 tokens/s` at 512 tokens;
- generation: `67.77 ± 0.09 tokens/s` at 128 tokens;
- full 128-sample LLM-planned pipeline: `1.007 s`;
- LLM and offline physical `runs`: identical.

The latest code gate for the merged multicomponent bridge is PR #9 run 30864728585: 356 passed, 15 skipped, workflow policy PASS, wheel PASS and whitespace PASS. The AMD ROCm evidence remains the dated 28 July core-platform snapshot at 231 passed with exact FP64 primary-GCS/platform parity; it predates the PHREEQC bridge merge and is not a full current-main rerun.

11. Verification

The dependency-light controller evidence contains 216 passing tests plus 13 optional-PyTorch skips; the PyTorch controller checkpoint passes 231 tests, and the retained AMD ROCm core-platform snapshot passes 231. The latest PR gate adds the PHREEQC multicomponent contracts and reports 356 passed/15 skipped. Together these checkpoints cover:

- physical formulas and analytical cases;
- K+ monotonicity and radioactive decay;
- non-finite, negative, missing and unknown inputs;
- strict site metadata;
- retrieval ranking and no-match behaviour;
- permission denial and recursive privacy redaction;
- private memory permissions and session isolation;
- deterministic and mock-LLM planning;

- artifact generation and checksum verification;
- localhost dashboard health, POST execution and artifact serving.

The PHREEQC tests establish contract/schema behaviour, dynamic selected-output closure, deterministic diagnostics, fail-closed execution and report artefact integration. They do not establish calibrated chemistry, external scientific agreement or regulatory validity.

12. Safety boundary

NuclidePath must not be used to:

- calculate public dose or regulatory compliance;
- declare a site or water source safe;
- issue protective-action advice or public alerts;
- control equipment or response systems;
- substitute for site measurements, calibrated models or qualified authorities.

The current screening model omits heterogeneous/transient flow, multidimensional transport, nonlinear/kinetic sorption, colloids, preferential pathways, matrix diffusion, daughter products and source-mass calibration.

13. Acceptance criteria

All project acceptance criteria are met when:

- offline and local-LLM demos complete from documented commands;
- all tests pass;
- all five Track 2 capability flags are true;
- the same case gives identical physical runs with both planners;
- source URLs, assumptions, classifications, warnings and uncertainty ranges appear in outputs;
- every artifact checksum verifies;
- the dashboard is usable on localhost;
- AMD runtime/model/hash/benchmark evidence is documented with its source revision and workload scope;
- the optional PHREEQC bridge is documented as process/comparative evidence only;
- README, specification, video script and slide deck are in English;
- public contest publication is not marked complete while the source fork is stale or the video URL is a placeholder.